



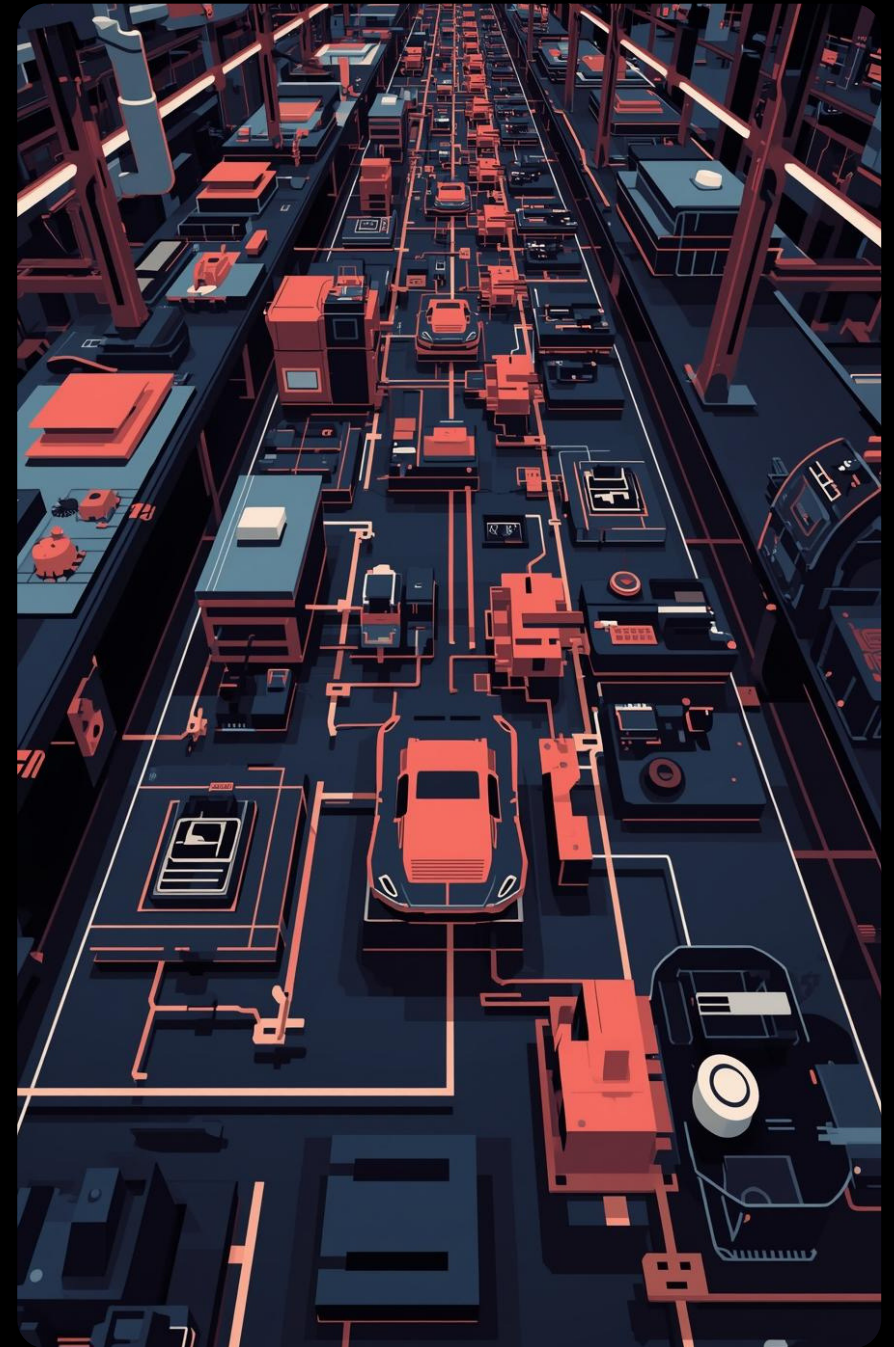
Massachusetts Institute of Technology

Mass Production of RC Car Manufacturing

2.854 Intro to Manufacturing Systems

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Background and Motivation

Optimization of RC Car Production System



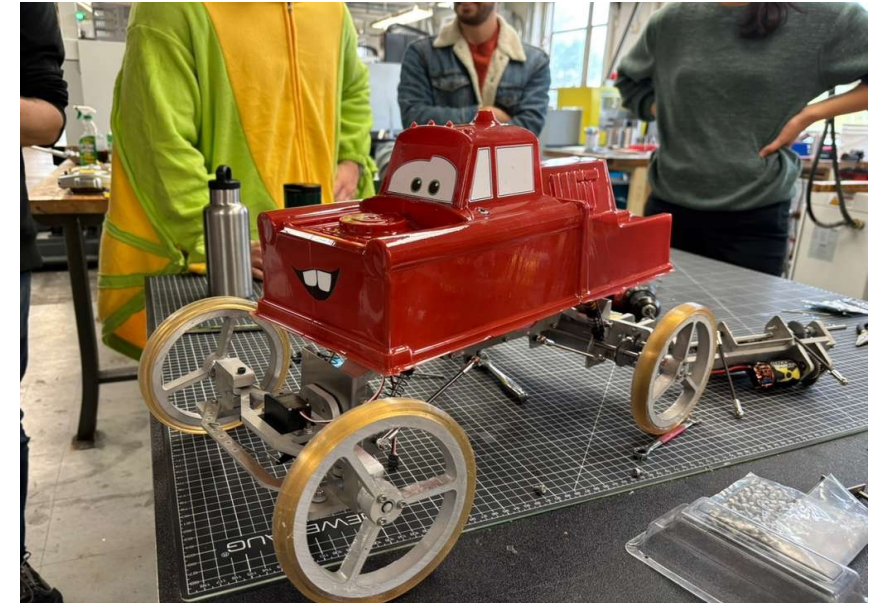
Premise & Background

- **2.810 - Design & Manufacturing of 40 RC cars** as part of a scaled production project
- Mirrors real-world subassembly manufacturing including **Drivetrain, Wheels, Chassis, Suspension, Steering**
- As 2.810 students, we saw major delays and machine bottlenecks even while building 40 units



Motivation

- Our drivetrain team struggled to produce just 40 units, raising the question: ***Is the machine shop truly optimized for all teams to work together efficiently?***
- Do we have the right number of machines and the right allocation of people? ***Is the waterjet actually the bottleneck, or is something else limiting throughput?***
- If this were a real RC-car company, how could we model and optimize MIT's machine shop for maximum output and profit under real constraints?



Provide Data-Informed Recommendations to LMP



Machine & Subteam Sizing, Partnership with other campus shops



Real Manufacturing Scaling → Capacity Planning & Bottlenecks

Guiding Questions & Introduction

Optimizing Production Flow, Machine Capacity, Operator Subteam Pools, and Profitability

Key Questions:

- 1) Given the **current equipment, labor, and layout in the LMP shop**, **how many RC cars can we produce in a year?**
- 2) How much could **production increase if the machine allocation, and operator assignments were optimized?**
- 3) Would **investments in new machines** or resource allocation be financially justified?



Factory Flow Modeling

Map drivetrain, chassis, suspension, wheels, and steering processes. Identify number of operators per team, cycle times, and machines per workstation.



Queueing & Utilization

Simulate and model the processes for all subassemblies to analyze operator and machine efficiency across teams.



Throughput Optimization

Keep WIP and queues under control in the RC car assembly by prioritizing tasks, purchasing new machines, and reallocating people to adequate subassemblies.



Economic Optimization

Identify the total costs to build an RC car and evaluate whether additional investments are justified based on the profit generated from throughput.

Objective, Constraints, & Assumptions

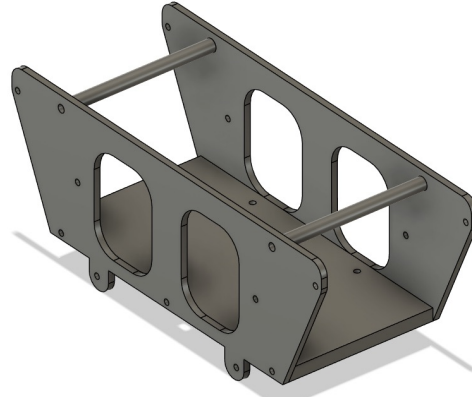
Objective: Maximize yearly throughput and profit by optimizing machine capacity, operator allocation, and factory flow.

RC Car Subassemblies Overview:

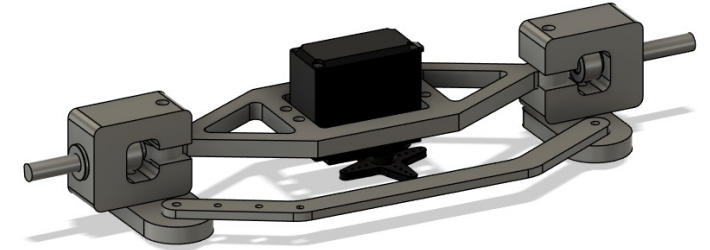
- Drivetrain (4 members)
- Chassis (4 members)
- Suspension (4 members)
- Steering (6 members)
- Wheels (6 members)

Key Assumptions/Constraints:

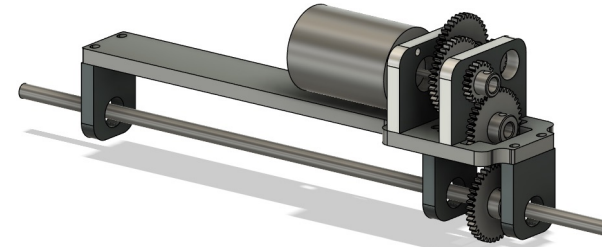
- Total operators cannot exceed 24 people
- Factory open 10 hours/day; **each subassembly team gets 2 hours/day.**
- Limited by existing LMP machines, layout, and space.
- Profit margin = 30%, with infinite arrival rate and demand.
- RC Car requires 1 Drivetrain + 1 Chassis + 1 Steering + 2 Suspension + 4 Wheels



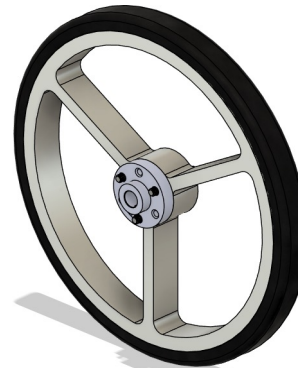
Chassis



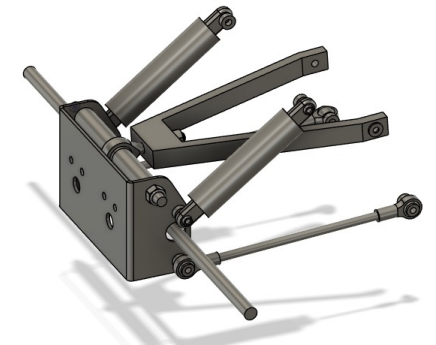
Steering



Drivetrain



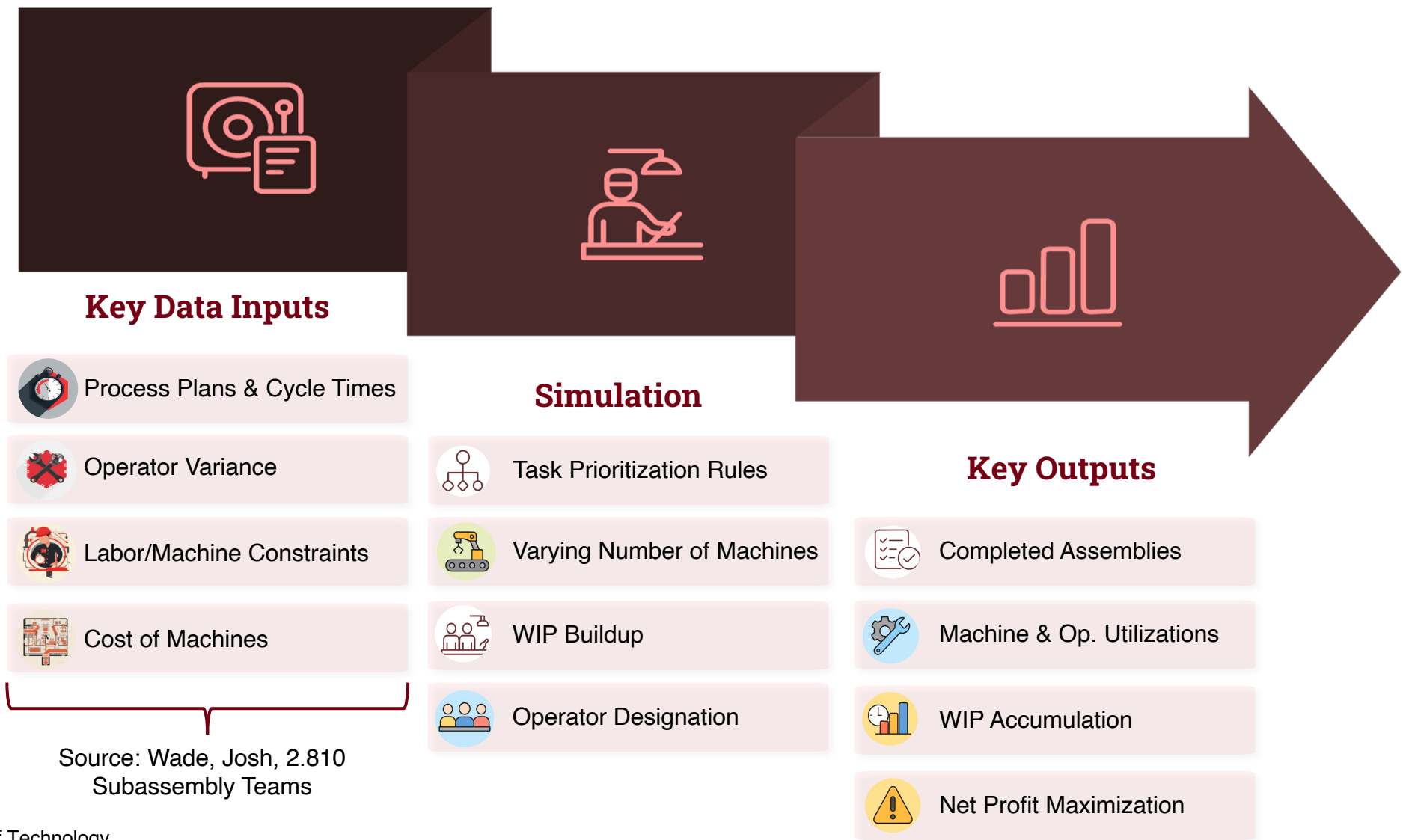
Wheels



Suspension

Characterization of Required Data

Optimizing Production Flow, Machine Capacity, Operator Subteam Pools, and Profitability



Data Acquisition, Challenges, & Analysis

Data collection enabled accurate modeling and bottleneck identification despite variability.



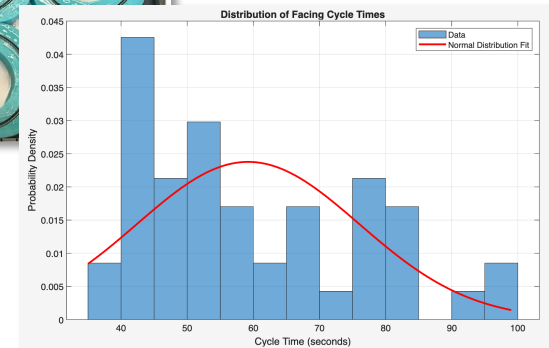
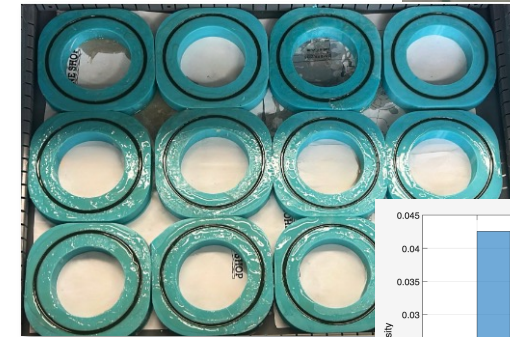
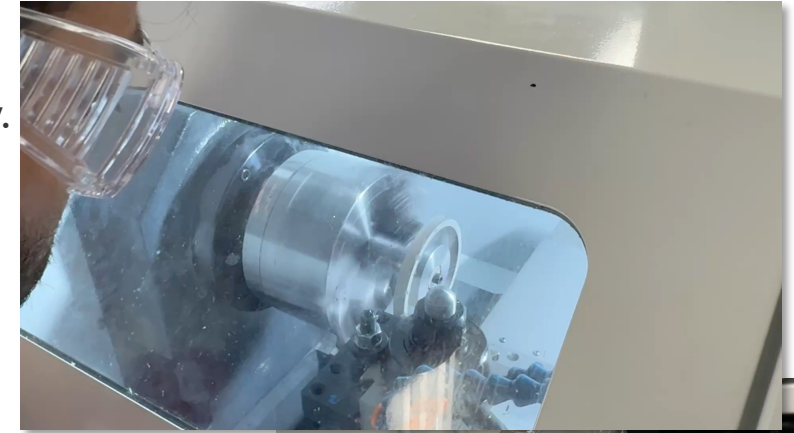
Data Acquisition

- **Process Mapping:** Shadowed each team to understand their mfg. process
- **Sampling:** 10+ samples of cycle time per process step to account for variability.
- **Labor Constraints:** 10-hour shop, 2-hour team windows, operator sharing.
- **Cost Analysis:** BOM review, \$203.53 production cost.
- **Observation:** Identified bottlenecks and inefficiencies in LMP shop.



Challenges Encountered

- **Variability:** Cycle time variability and operator performance inconsistency (Accounted for multiple samples).
- **Processing Complexity:** Batch vs. individual part processing complexity.
- **Baseline Model Validation:** Verify model results match intuition and actual shop floor constraints.



System Modeling Method/Considerations

Our team evaluated several modeling and problem-solving approaches before selecting the method most capable of capturing the complexity of the real production environment.

Requirements

- **Factory Flow:** Model full end-to-end RC car production flow
- **Utilization:** Capture machine/operator utilization & queues
- **Variability:** Cycle time variability & change operator/machine count
- **Flexibility:** Test alternative resource allocation & layouts
- **Analytics:** Output scaling and profit analysis
- **Visualization:** Generate visual comparisons & plots

Tools Comparison

Excel/Spreadsheets

- ✓ Easy calculations
- ✗ Cannot model WIP/variability

Python/MATLAB

- ✓ Flexible, plots, scaling
- ✗ Not inherently DES, Time-Consuming, Limited Test Options

Machine Learning

- ✓ Predictive
- ✗ Needs large data, ignores process flow

AnyLogic/DES Simulations

- ✓ Models queues, WIP, interactions
- ✗ Steep Learning curve

Digital Twin

- ✓ High-fidelity
- ✗ Complex setup, overkill

Selected Approach

Primary Solution

- **AnyLogic (DES):** Can simulate flows, queues, WIP, bottlenecks, schedules

Secondary Support

- **Python:** Used for post-processing, scaling, plots, profit analysis

This hybrid strategy leverages the strengths of both tools for comprehensive analysis.

AnyLogic

DES Capabilities

AnyLogic excels in discrete event simulation, capturing complex interactions among processes, queues, and constraints effectively, surpassing traditional spreadsheets.

Visualization

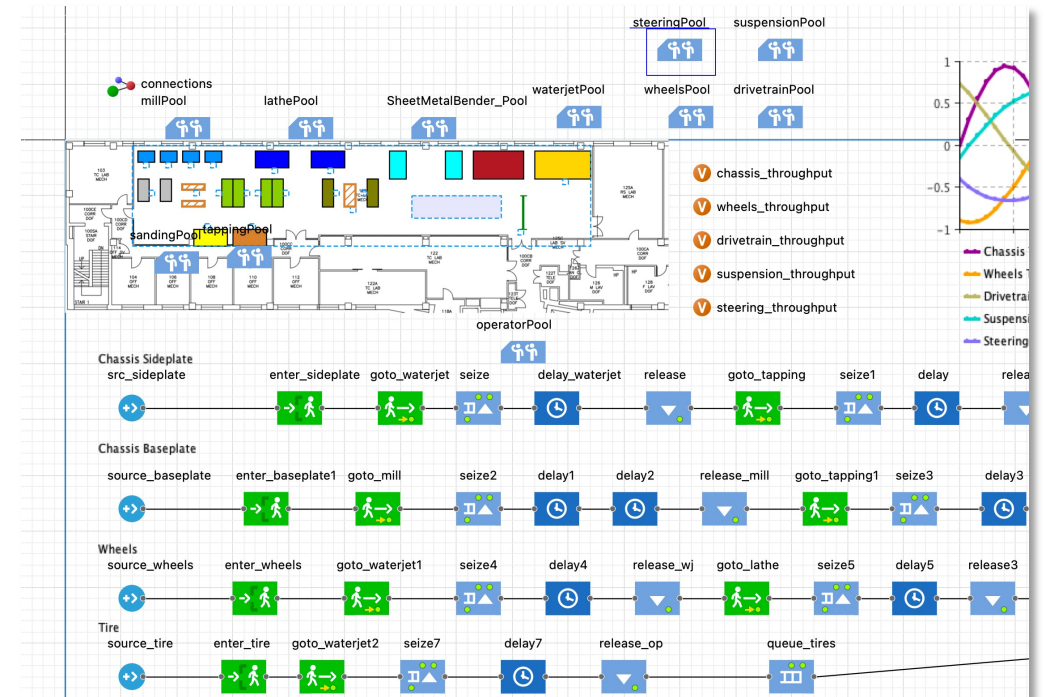
It provides compelling visual simulations for better validation and communication, helping stakeholders understand complex dynamics intuitively.

Layout Modeling

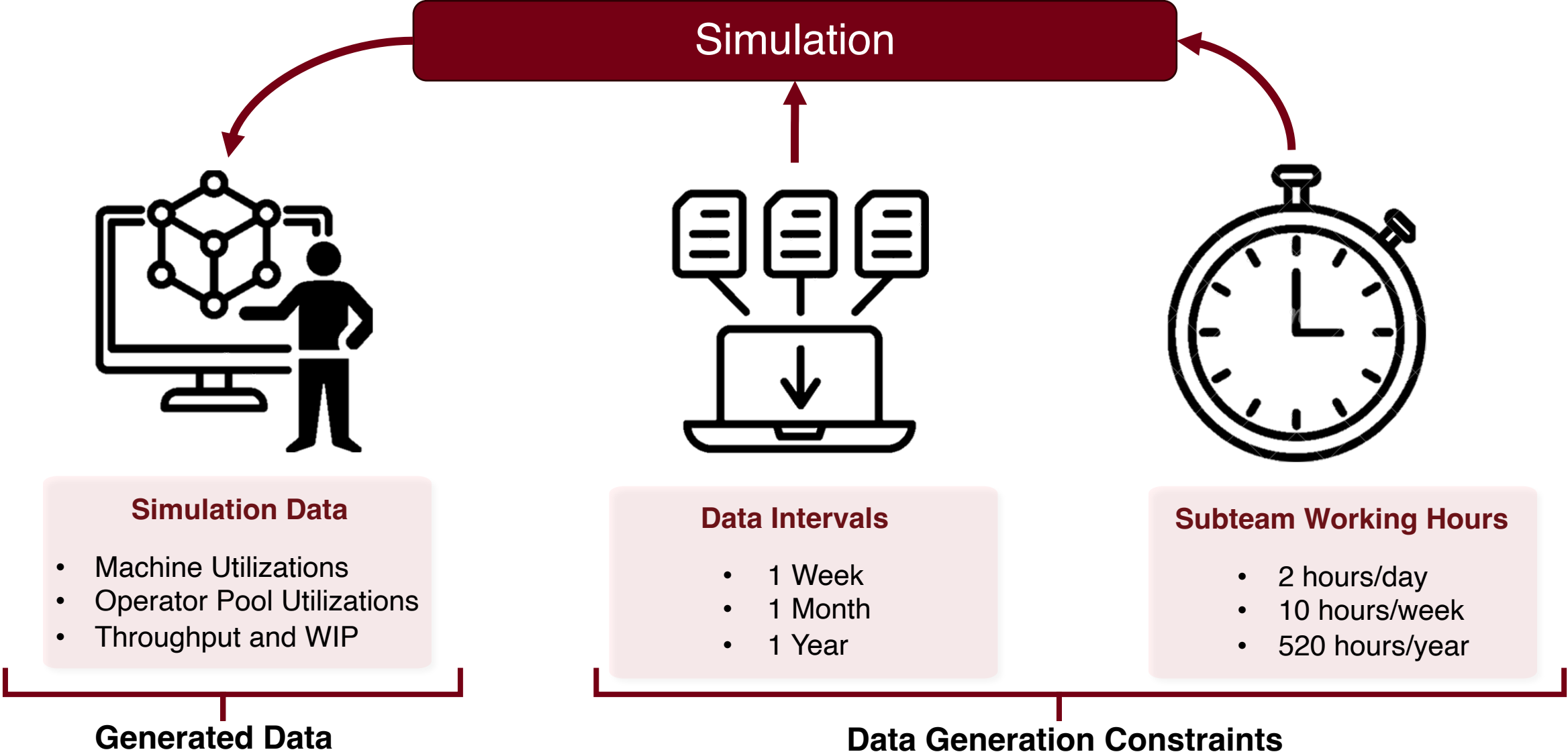
The software supports real layout modeling, by importing a .png files of the floor plan and enabling precise placement of workstations and analysis of travel distances.

Accessibility

The software is free to use compared to others like Siemens PlantSim (it is also compatible with Mac devices!).

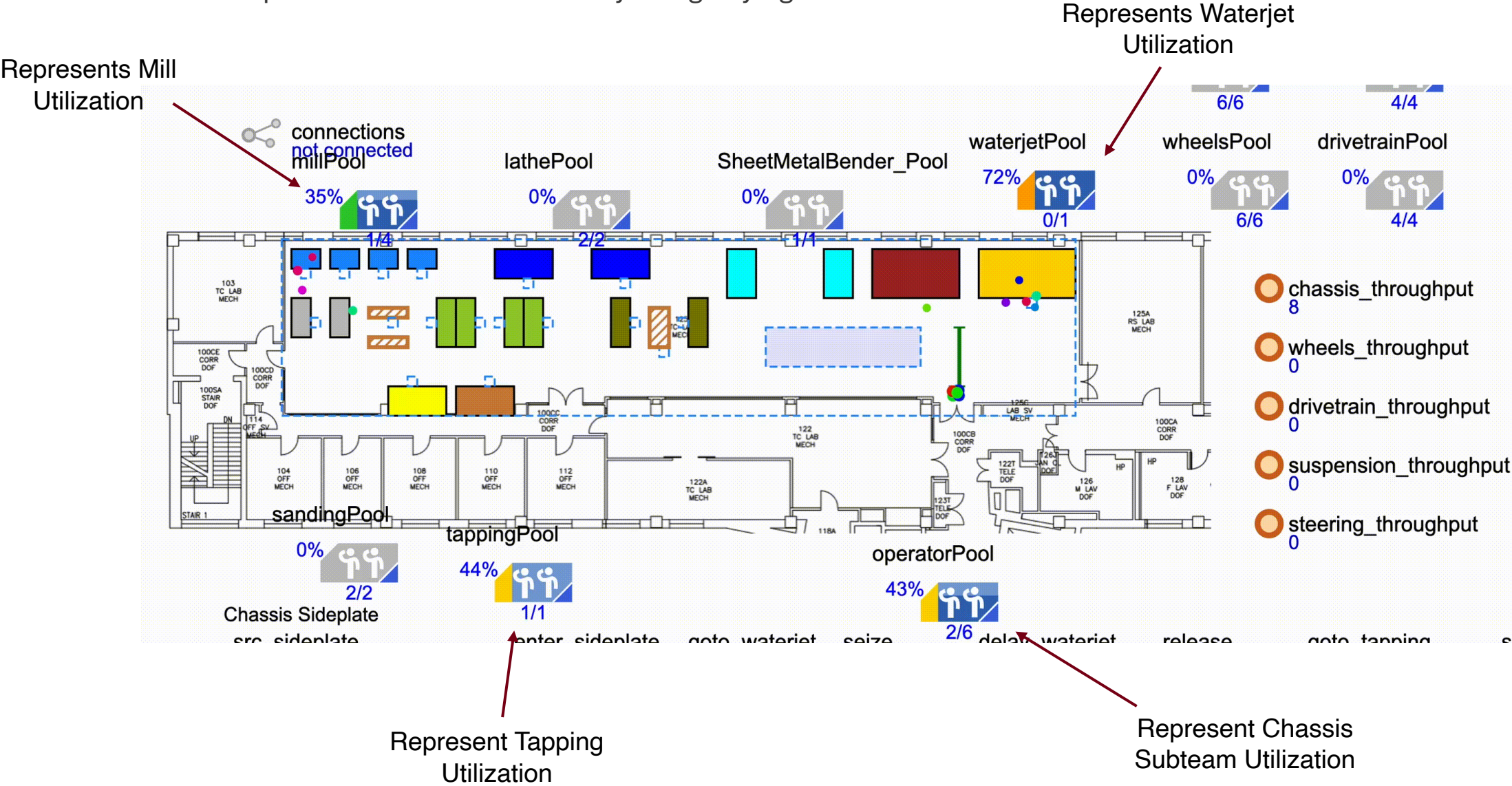


Results – Simulation Methods



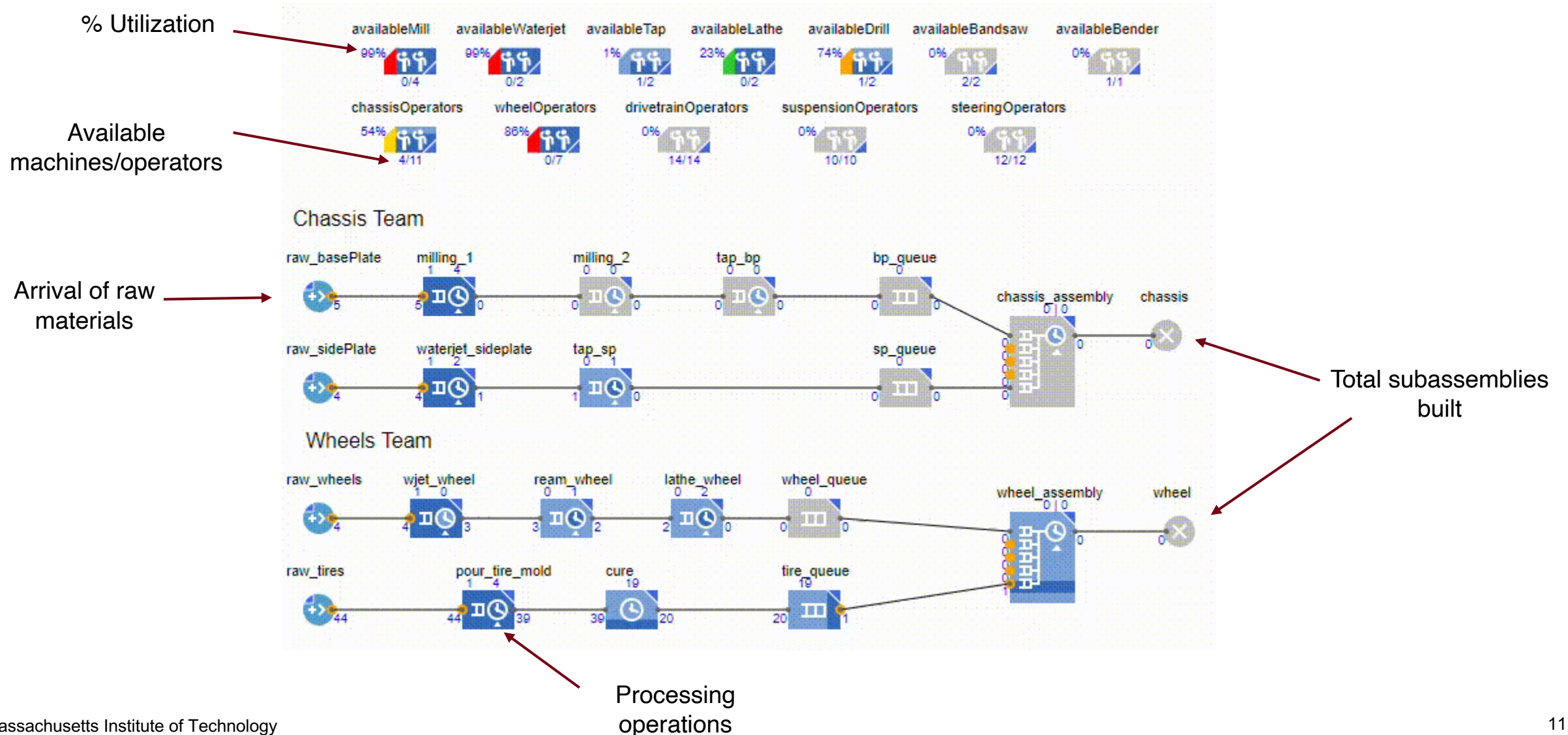
Results – Simulation

Simulation Model Example for Chassis Subassembly using Anylogic

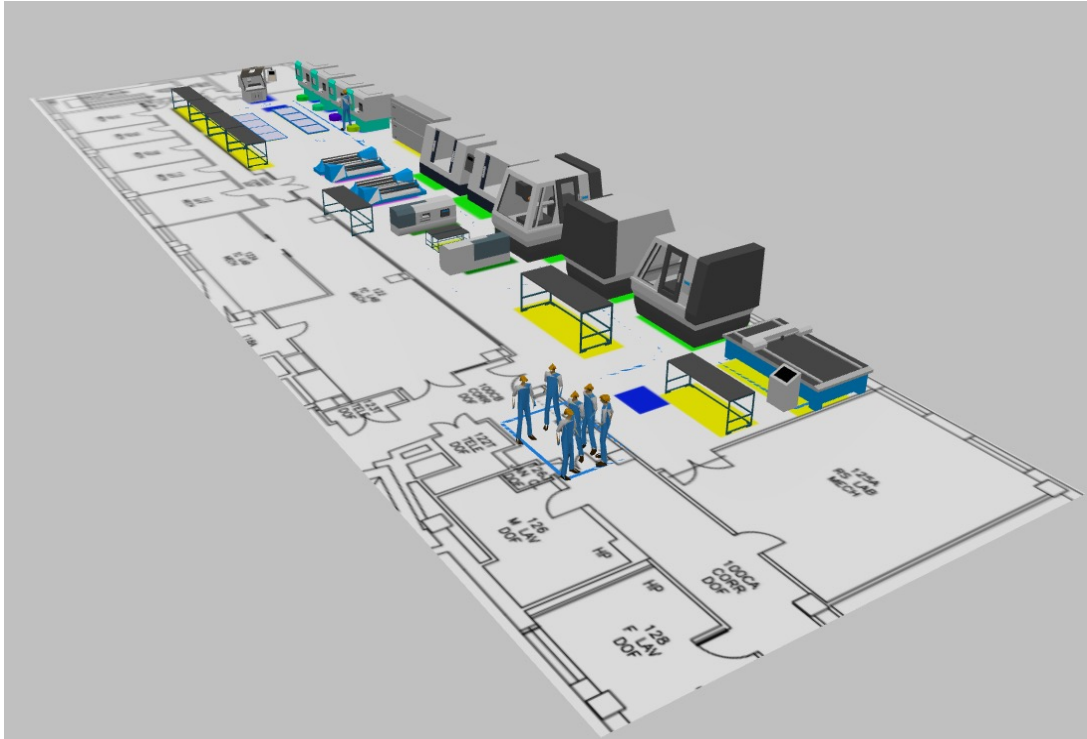


Results – Simulation

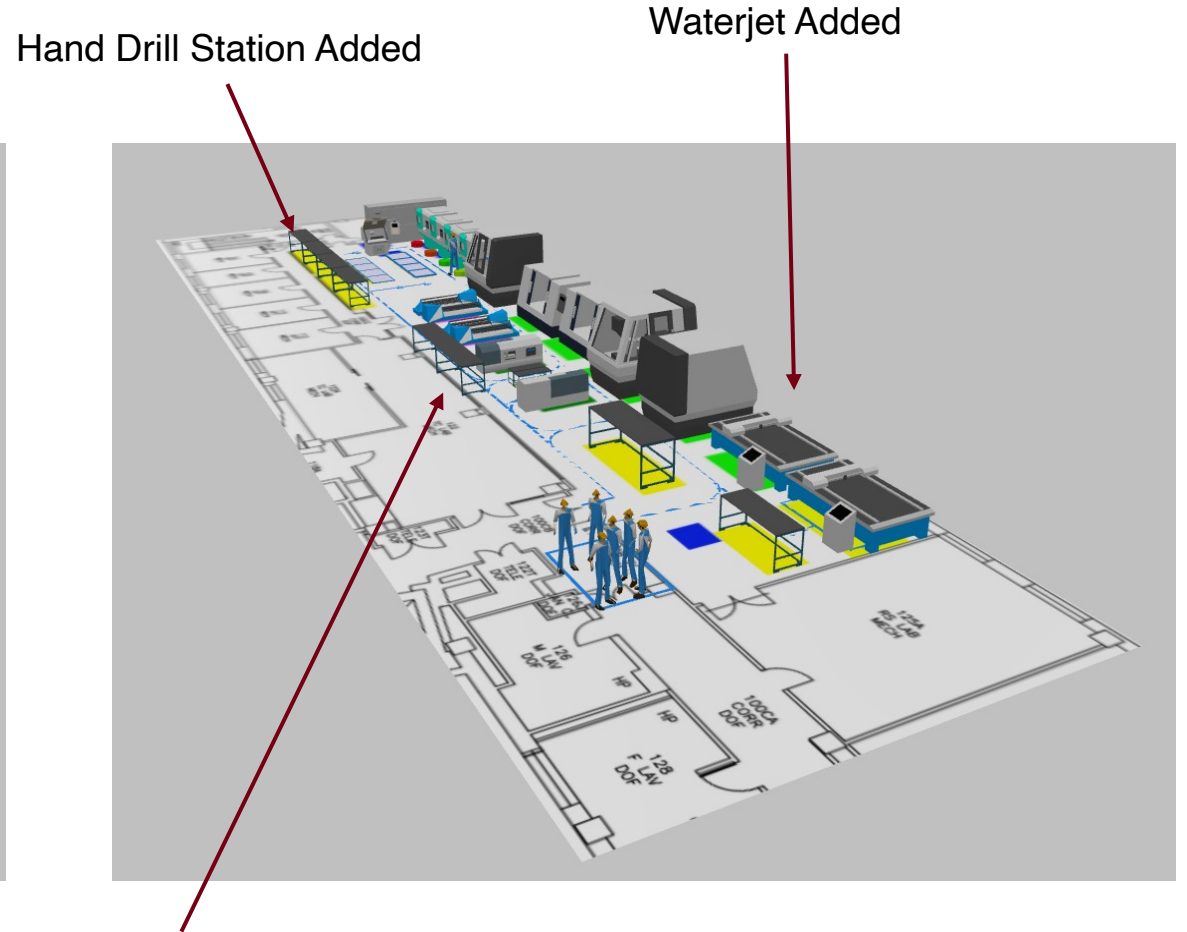
Simulation Model Example for Chassis and Wheels Subassembly using Anylogic



3D Simulation Modeling Layout



Old LMP Layout

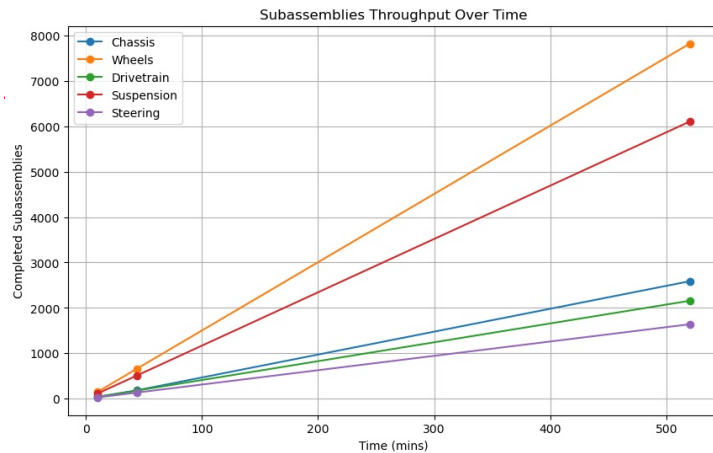


Tapping Station Added

Optimized LMP Layout

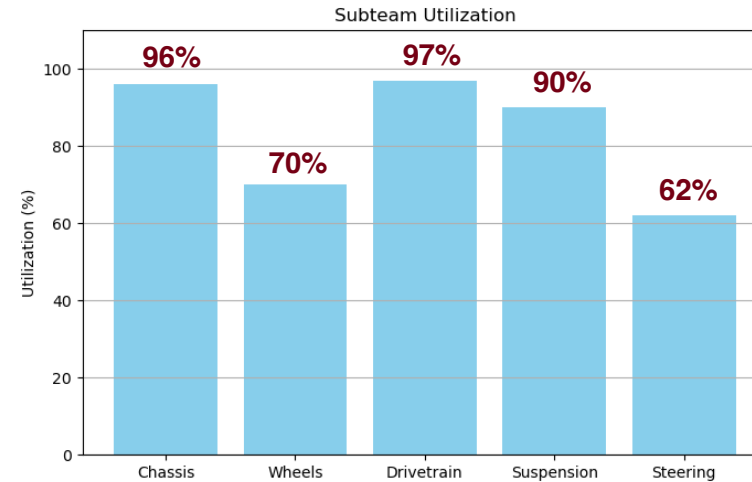
Results – Simulation Results & Key Outputs

Throughput of Subassemblies



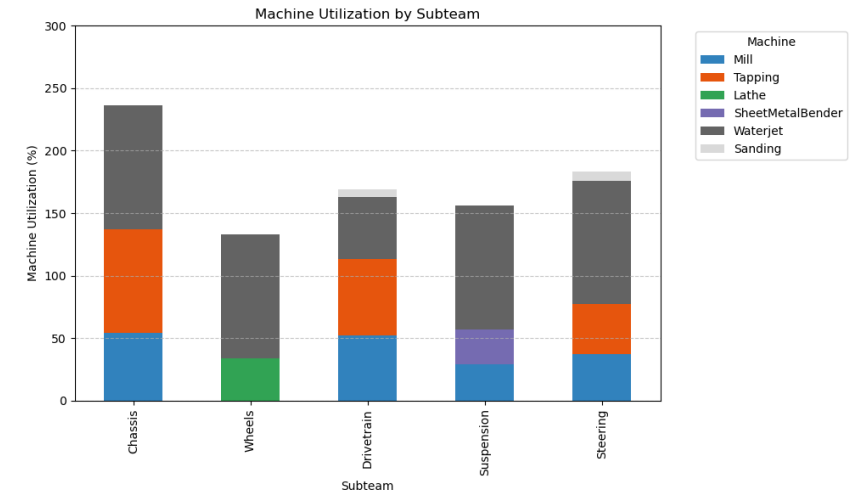
- Suspension has the highest throughput to make most RC Cars in a year
- Steering is the bottleneck; Car production only improves by fixing it
- **Variability in subassembly throughput shows inefficiency**

Subteam Utilization



- Drivetrain subassembly is bottlenecked by their lack of members
- Steering limits overall production and has low member utilization due to machine bottleneck

Machine Utilization



- Waterjet is the main bottleneck; needed by all teams and most utilized.
- Tapping limits Drivetrain; Milling usage high but sufficient capacity
- Hand-Drill operation affects Wheels significantly as well

Results – Analysis of Primary Bottleneck Machine

1

Raw Utilization

Measures how busy each machine is across subteams.

2

Bottleneck Score

Normalizes for production output.

3

High Score Impact

Highlights limiting machines per unit produced, even with high utilization.

- **Waterjet** is the dominant bottleneck, constraining 4/5 of the subteams.
- ***Steering has the highest bottleneck score***, making it the largest limiter of complete RC car throughput.
- Drivetrain's unique bottleneck is the tapping station, not the waterjet.
- Bottleneck scores help quantify where machine investments (e.g., additional Waterjet or tapping capacity) yield the best system-wide improvement.

** Bottleneck score = machine utilization ÷ subteam throughput*

Subteam	Primary Bottleneck Machine	Bottleneck Score
Chassis	Waterjet	0.038268
Wheels	Waterjet	0.012670
Drivetrain	Tapping	0.028280
Suspension	Waterjet	0.016230
Steering	Waterjet	0.060366

Results – Bottleneck Analysis & Optimization

Increase steering throughput by additional machines and operators to increase overall RC Car throughput

1

Steering is the bottleneck:

Improving steering throughput is critical to increasing overall RC Car production

2

Adding a Waterjet:

To alleviate a 99% WJ utilization bottleneck for steering, we added a waterjet which increased throughput and net profit

3

Adding a team member:

Number of people became bottleneck after adding a WJ, added one member (6 → 7 people), further increasing throughput and net profit

4

Limit Reached:

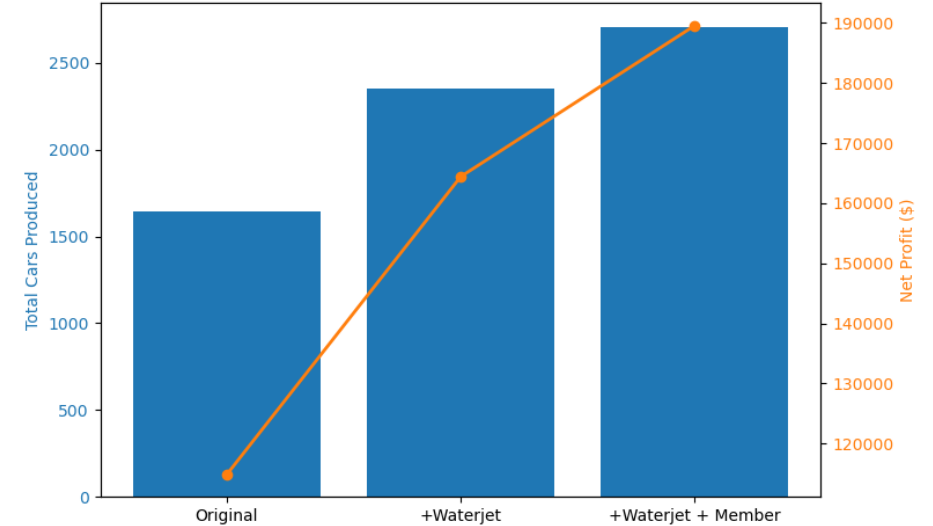
After these improvements, machines and/or people were no longer over-utilized, indicating that production is now time limited.

5

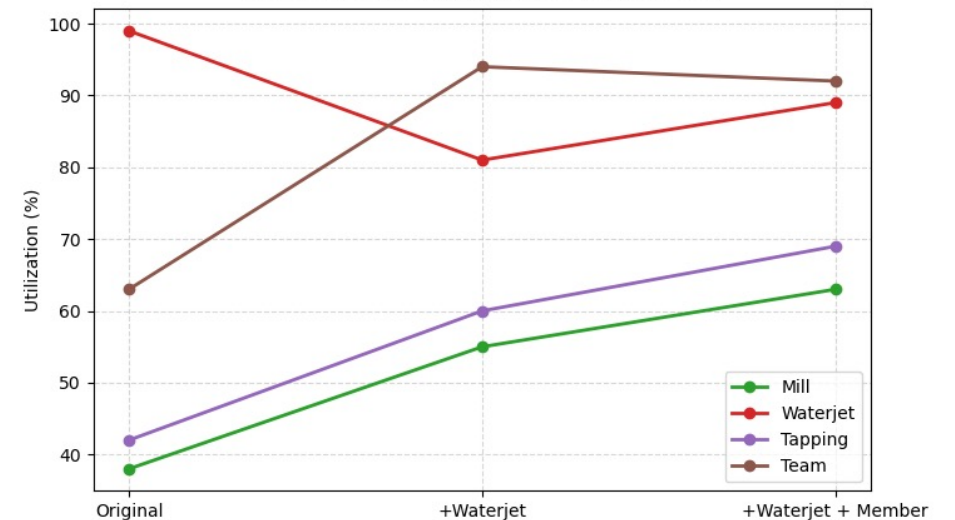
New Steering Subassemblies:

2708 steering subassemblies were made after adding 1 WJ and 1 person. While NOT the most optimized version, it helps increase throughput

Effect of Steering Optimization on Throughput and Profit



Machine & Team Utilization Across Scenarios



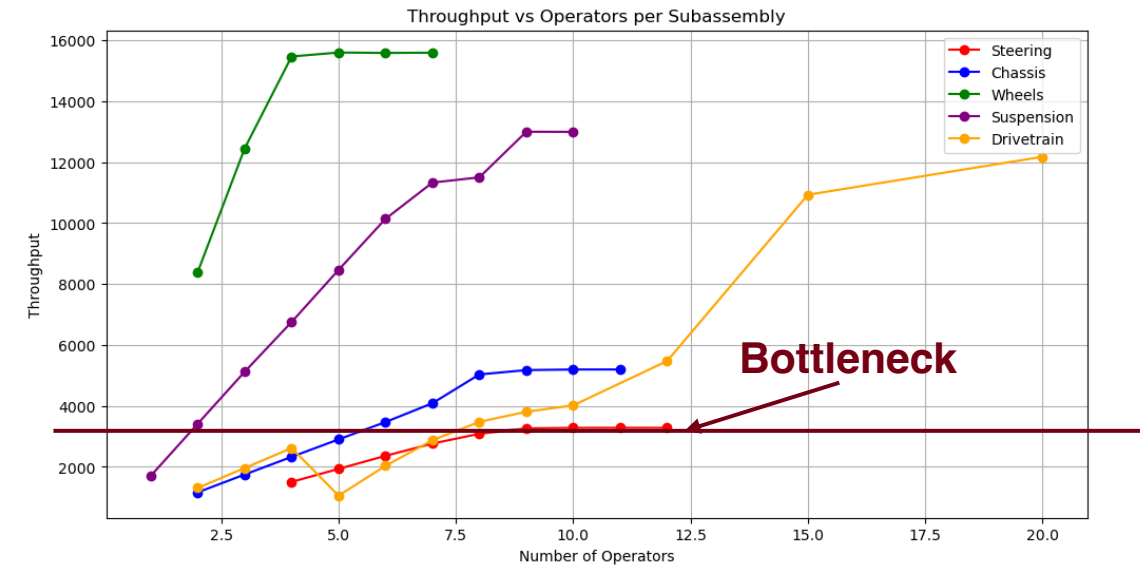
Results – Optimal Operator Allocations

Increase steering throughput by additional machines and operators to increase overall RC Car throughput

Optimizing Machine Allocations

- We applied the same optimization approach to Wheels, Drivetrain, Chassis, and Suspension to reduce bottleneck machines.
- **Added 1 Tap, 1 Waterjet, and 1 Hand Drill/Reamer** across the shop.
- These changes lower machine utilization across the shop
- The new focus is on **optimizing operator allocation** for each subteam to match steering throughput.

Machine	Count
Mill	4
Waterjet	2
Tap	2
Lathe	2
Reamer / Hand Drill	2
Bandsaw	2
Sheet Metal Bender	1



Optimizing Operator Allocations

- Swept through multiple operator allocations using AnyLogic simulation to find throughput per subassembly.
- With 24 total operators and steering as the bottleneck, optimized allocation.
- Achieved **maximum throughput of 2,765 parts/year**, with all subassemblies producing near this level.

Costs

Cost of a Car

Subassembly*	Cost
Steering	\$25.48
Suspension	\$55.96
Chassis	\$16.75
Drivetrain	\$66.11
Wheels	\$39.23
TOTAL Manufacturing Cost	\$203.53
Profit Margin	30%
TOTAL Profit Margin	\$61.06

* Detailed breakdown of subassembly costs in Appendix B

Cost of Machines

Machine	Price
Manual Mill (Bridgeport)	\$17,000
Waterjet Cutter (OMAX 2652)	\$27,500
CNC Lathe (HAAS TL-1)	\$35,995
Hand Drill	\$200
Bandsaw (DoALL Vertical Bandsaw)	\$17,000
Sander (JET Vertical Combo)	\$1,649
Tapping	\$1,100

- Total manufacturing cost of a car is **\$203.53**. The RC car industry typically experiences a profit margin ranging from 15% to 45%. Taking the average profit margin (30%) results in a total sale price of **\$264.59** (a profit of **\$61.06**).
- To optimize production, additional machines may need to be purchased. The Cost of Machines table outlines the prices of each piece of machinery. Purchasing a machine will reduce the total profit for the year and must be offset by a corresponding amount of sales to reach a breakeven point.

Profit Analysis

Unmodified Production Line

The unmodified production line features the same quantity of machinery available as the current LMP shop.

Profit Generated

Total # of cars produced = 1640

Profit generated = $\$61.06 \times 1640 = \mathbf{\$100,138.40}$

Optimized Production Line

This production line adds an additional waterjet, hand drill, and tap to maximize the throughput.

Profit Generated

Total # of cars produced = 2765

Gross profit generated = $\$61.06 \times 2765 = \$168,830.90$

Net profit = $\$168,830.90 - \$27,500 - \$1,100 - \$200 = \mathbf{\$140,030.90}$

Profit gain = $\$140,030.90 - \$100,138.40 = \mathbf{\$39,892.50}$

Machine	Qty
Mill	4
Waterjet	1
Tap	1
Lathe	2
Hand drill	1
Bandsaw	2
Bender	1

Machine	Qty	Delta
Mill	4	0
Waterjet	2	+1
Tap	2	+1
Lathe	2	0
Hand drill	2	+1
Bandsaw	2	0
Bender	1	0

Comparison to Real World Data and Methods

Model Highlights 	Discrepancies 
Waterjet identified as bottleneck (prior experience)	Operator variability not fully captured: model assumes perfect workflow and consistency
Tapping Delays matched real-world congestion	Setup time, homing waterjet, tool changes, etc., not included within current model
Throughput ranking of subassemblies aligned with real workflow observations	Shop layout not completely accurate
Model reproduced the same difficulty producing 40 cars, validating queue buildup and machine competition	Reliability data was not account for – machines had incomplete MTTF/MTTR data logs from LMP shop

Future Work

Data Expansion/ Accuracy

- Collect more cycle times with different operators.
- Understand each team's priority queues to mimic current scenario.

Model Enhancement

- Enhance models by adding more complexity to mimic real-life scenarios:
- Batch processing in the waterjet.
 - Set-up times during team switches.
 - Allocate lead time for stock material purchases.
 - Include MTTR/MTTF into the model

Improved Optimization

- Optimize operation orders to produce parts more efficiently.
- Model how different designs, materials, manufacturing methods will affect throughput and cost.

Conclusion

Current Setup

- *Estimated Maximum Throughput: 1,640 cars per year*
- *Estimated Net Profit: \$100,138.40*

Final Results

- Maximum Throughput: **2,765 parts per year**
- Optimized Profit: **\$140,030.90**
- **Net Profit Difference: \$39,892.50**

Shop Layout

- 4 Mills
- • 2 Waterjets (+1)
- • 2 Taps (+1)
- 2 Lathes
- • 2 Reamers (+1)
- 2 Bandsaws
- 1 Bender

Operators per Team

- Drivetrain: 7 operators (+3)
- Steering: 7 operators (+1)
- Chassis: 5 operators (+1)
- Wheels: 3 operators (-3)
- Suspension: 2 operators (-2)

Recommendations to LMP Shop Staff

1. Waterjet is a significant bottleneck for multiple teams
 - LMP should partner with another lab with a waterjet to dissipate demand
 - Purchasing and installing a new waterjet with current LMP setup may not be feasible
2. Consider student allocation (operators per team) for each subteam to maximize efficiency for 2.810 course

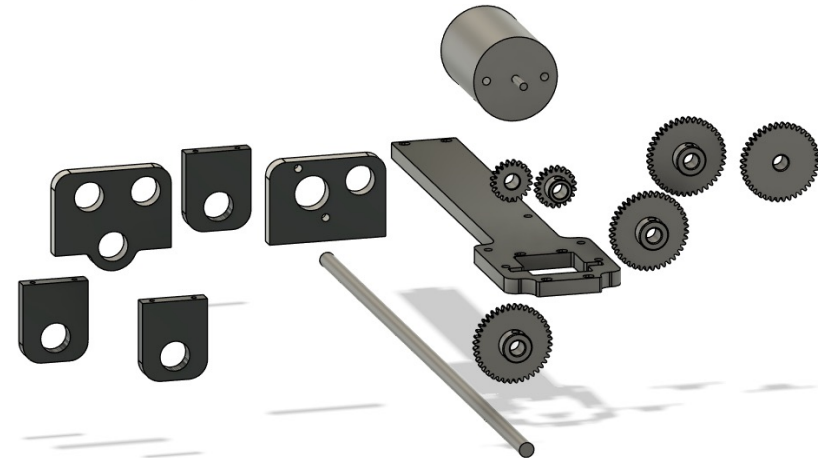
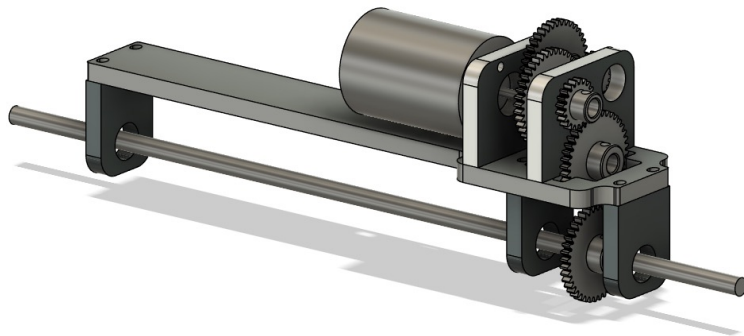
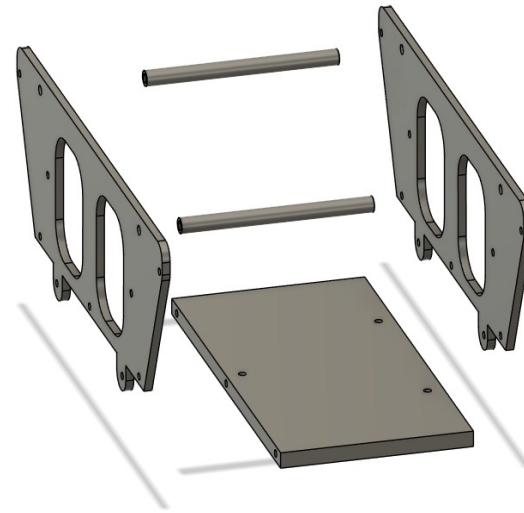
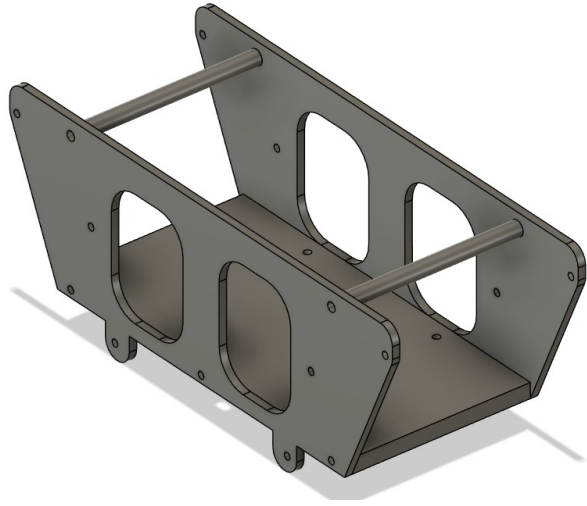


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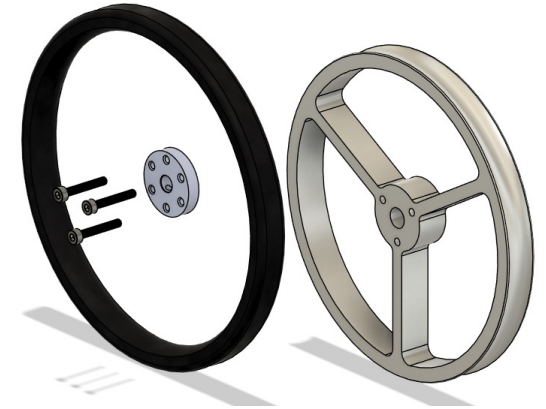
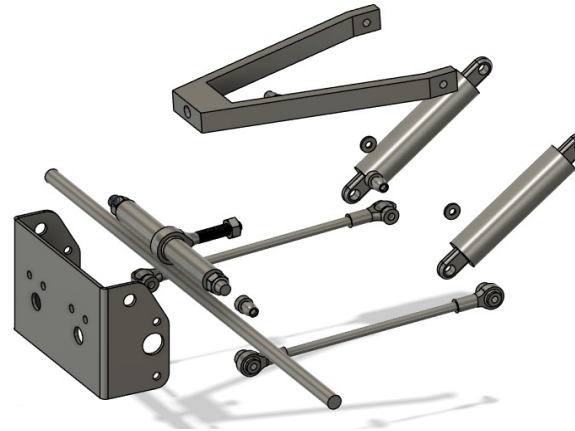
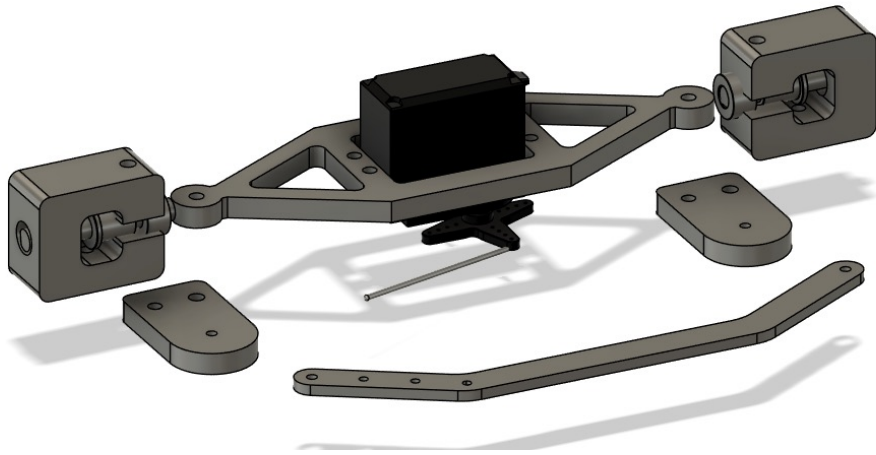
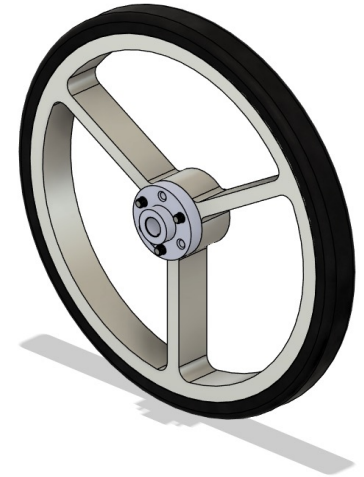
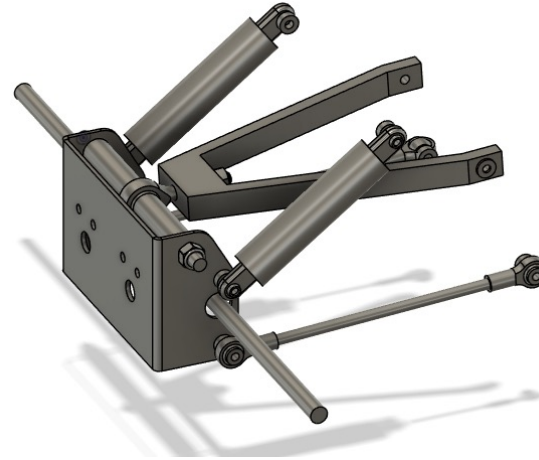
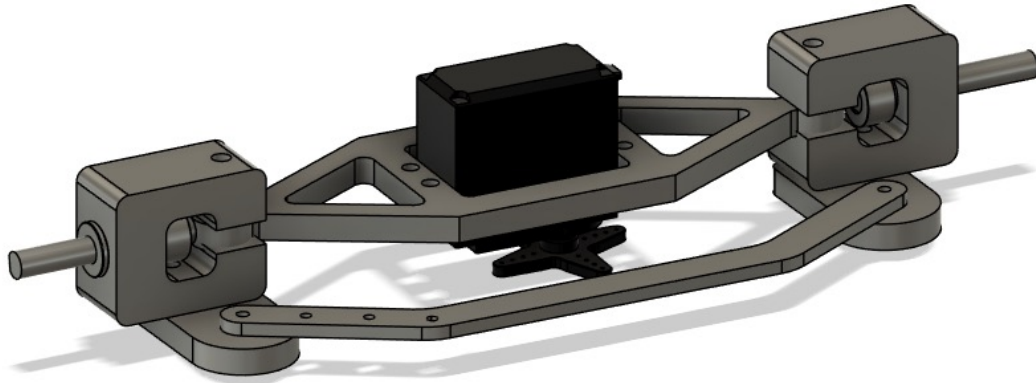
Thank You!

Any Questions?

Appendix A: CAD Models of Subassemblies



Appendix A: CAD Models of Subassemblies



Appendix B – Bill of Materials (Steering)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Rigid Plate Mount	12.8	0.25	1	\$2.75	
Rigid Linkage	8.353338	0.125	1	\$1.26	
C-Joint	1.875	0.75	2	\$1.83	
Steer Bars	1.75	0.25	2	\$0.75	
M4 Nylon Washers			5	\$0.40	\$0.08
Bushing			2	\$5.90	\$2.95
Shaft Collar			2	\$3.70	\$1.85
M4 45mm Screw			2	\$0.60	\$0.30
M4 Locknut			2	\$0.28	\$0.14

Appendix B – Bill of Materials (Steering cont.d)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
M4 14mm			2	\$0.28	\$0.14
M4 Nut			2	\$0.08	\$0.04
M3 16mm Screw			2	\$0.24	\$0.12
M3 12mm Screw			2	\$0.24	\$0.12
M3 Locknut			4	\$0.16	\$0.04
M3 10mm Screw			4	\$0.52	\$0.13
M3 8mm Screw			2	\$0.26	\$0.13
0.25" Ø D-Rod	1.5		2	\$1.97	
1/16" Ø Steel Wire			1	0	
Servo Motor			1	\$4.25	\$4.25

Total cost = \$25.48

Appendix B – Bill of Materials (Suspension)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Ball Joint			2	\$4.50	\$2.25
Connecting Rod (M5 x 100 mm)			2	\$1.70	\$0.85
M5 Spacers 35mm			4	\$1.98	\$0.49
M5 Lock nuts			6	\$0.33	\$0.05
M3 Lock Nut			20	\$0.83	\$0.04
M3 Washer			4	\$0.06	\$0.02
M3 Bolt 18 mm			4	\$0.24	\$0.06

Appendix B – Bill of Materials (Suspension contd.)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Wishbone	12.3516	0.375	2	\$7.28	
C-Channel	11.066625	0.125	2	\$3.35	
M4 Ball Joint Caps - Metal + Threaded			8	\$8.00	\$1.00
M4 Threaded Rods - 2m McMaster			4	\$5.32	\$1.33
M3 Bolts 20mm			16	\$0.96	\$0.06
M3 Spacers			16	\$0.42	\$0.03
RC Shocks 100 mm			4	\$20.99	\$5.25

Total cost = \$55.96

Appendix B – Bill of Materials (Chassis)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Base Plate	18	0.25	1	\$3.87	
Braces (Aluminum 1/4" OD, 3")			2	\$4.60	\$2.30
M3 Screws (M3 x 0.35 mm Thread, 6 mm Long)			6	\$0.54	\$0.09
M4 Screws (M4x0.5mm, 8mm length)			6	\$0.48	\$0.08
Side Plates	24	0.125	2	\$7.27	

Total cost = \$16.75

Appendix B – Bill of Materials (Drivetrain)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Base	9.73	0.25	1	\$2.09	
Axle Holder	2.18	0.25	3	\$1.40	
Gear Block	2.27	0.25	1	\$0.49	
Motor Mount	1.23	0.25	1	\$0.26	
12 Tooth Gear	0.3		1	\$1.73	
20 Tooth Gear	0.375		2	\$3.70	
40 Tooth Gear	0.375		4	\$11.07	
Short axle (26.4mm)	1.039		2	\$1.36	
Medium axle (33.25mm)	1.3090551		1	\$0.86	
Drive Axle (304mm)	11.9685		1	\$7.86	

Appendix B – Bill of Materials (Drivetrain contd.)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Bearings			8	\$6.39	\$ 0.80
Shaft Collar			3	\$5.49	\$ 1.83
10-32 Set Screw			8	\$0.67	\$ 0.08
4-40 x 1/8 Set Screw			1	\$0.09	\$ 0.09
4-40 x 1/8 Cone Point Set Screw			1	\$0.25	\$ 0.25
4-40 x 1/2 Screw			14	\$0.86	\$ 0.06
Bullet connectors			4	\$0.36	\$ 0.09
16 AWG Wire			8	\$0.17	\$ 0.02
DC Brushed Motor			1	\$20.99	\$20.99

Total cost = \$66.11

Appendix B – Bill of Materials (Wheels)

Part	Area (in2)	Thickness (in)	Qty per car	Cost	Cost per Unit
Wheel Aluminum	9		4	\$12.13	
50A Tire	2.04		4	\$4.20	\$0.52
M3 Stainless Steel			12	\$0.98	\$0.08
Pololu Mounting Hub			4	\$21.92	\$5.48
				\$39.23	

Total cost = \$39.23

Appendix C – Full Simulation Model

